The empirical recurrence for OEIS A282393, proved by transfer matrix

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Abstract

OEIS A282393 counts $n \times 2$ arrays over $\{0, \dots, 1\}$ subject to a condition imposed on every CELL through its king-move neighbourhood, and it carries an empirical recurrence of order 6 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. A cell's neighbourhood meets three consecutive rows, so the state of a transfer matrix is a pair of consecutive rows and each step settles the middle one; an array is a walk, and $a(n)$ is a walk count on $S = 20$ vertices, which is C-finite. Whether the conjectured recurrence annihilates it is then decided by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A282393 is "Number of nX2 0..1 arrays with no 1 equal to more than three of its king-move neighbors.". It has offset 1 and begins

$$4, 16, 57, 213, 796, 2964, 11049, 41193, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 3*a(n-1) +a(n-2) +7*a(n-3) -2*a(n-4) -4*a(n-6).

As of the "Last modified" line on the live entry (Feb 28 2017 04:01:38, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 Cells, not blocks

The king-move neighbours of a cell are the (at most) eight cells surrounding it. Every cell carries the condition

every 1 has at most 3 neighbouring 1s,

and the count of neighbours enters it, so the boundary is not a detail that can be papered over: a corner cell has three neighbours, an edge cell five, an interior cell eight, and a condition such as "a strict majority of its neighbours" means different things at each. In particular one cannot pad the array with a border of zeros — that would give boundary cells eight neighbours and change what is being counted. The outside is therefore carried explicitly, as a sentinel row above the first and below the last and as a sentinel column either side, and a neighbour that falls outside is simply absent from both counts.

Write an array as a sequence of rows $u_1, \dots, u_L$, each in $\{0, \dots, 1\}^2$. The neighbourhood of a cell in row u_i meets u_{i-1}, u_i and u_{i+1} and nothing else. Take as vertices the pairs $(r, s) \in$

$(\{0,\ldots,1\}^2 \cup \{\varnothing\}) \times \{0,\ldots,1\}^2$, and put an edge $(r,s) \to (s,t)$ when row s satisfies the condition at every one of its 2 cells, given r above and t below. An array with L rows is then a walk of length $L-1$ starting at $(\varnothing, u_1)$, each step settling one row, and the last row is settled by the terminal weight, which evaluates it with $\varnothing$ below. Hence, with M the adjacency matrix, ι the starting vector and τ the terminal weight,

$$a(n) \;=\; \iota^\top M^{n-1} \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C-finite.

3 The proof

Lemma 1. *Let $q(t) = t^6 - \sum_i c_i t^{6-i}$ be the characteristic polynomial of the conjectured recurrence. Then the residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j; and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+6} - \sum_i c_i M^{j+6-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ satisfy the monic linear recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; 3\,a(n-1) + a(n-2) + 7\,a(n-3) - 2\,a(n-4) - 4\,a(n-6)$$

for every $n > 6$.

Proof. The vector $w = q(M)\tau$ was formed by 6 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly. Those integers vanish from the index corresponding to $n = 6 + 1$ onwards, and the run was continued until the working vector was identically zero, which settles all larger j at once. By Lemma 1 the recurrence holds for every $n > 6$. □

4 Verification

Three checks, each independent of the algebra above.

First, the transfer model reproduces all 24 terms the entry publishes, exactly, in integer arithmetic. That check earned its place here. An early version of the construction tested whether a neighbour lay in the current row by comparing the row objects rather than their positions; when the row above happened to be identical to the current one, the cell directly above was then skipped as though it were the cell itself. The counts agreed with the entry for arrays of up to three rows and diverged at four, which is exactly the sort of error that a check on a handful of small cases would have missed.

Second, the conjectured recurrence was evaluated directly on the entry's own published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A282393.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.